SPARKY'S SCIENCE FACTS

Why Do We Have Day, Night and Seasons?

Astronomy · Ages 7+

> **Short answer:** We have day and night because the Earth spins. We have seasons because the Earth is tilted as it travels around the Sun.

How it works

- The Earth spins around once every 24 hours. The side facing the Sun has day; the other side has night.
- The Earth takes one year, about 365 days, to go all the way around the Sun.
- The Earth is tilted. When the southern half leans towards the Sun, sunlight is stronger and days are longer: it is summer in South Africa.
- Six months later the southern half leans away from the Sun, and we have winter.

Key words: Axis: the line the Earth spins around; Tilt: the lean of the Earth's axis

Common mix-ups

- Summer is not because the Earth is closer to the Sun. The Earth is actually closest to the Sun in January, which is summer here but winter in Europe.
- The Sun does not move around the Earth: the Earth spins, so the Sun seems to rise and set.

Did you know? In South Africa, Christmas is in summer, while in countries in the north it is in winter.

Try it: Shine a torch on a ball with a sticker for your home, and spin the ball slowly to see day and night come and go.

Ask Sparky: Why are winter days shorter? Do other planets have seasons?